# Medha Agarwal

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## Education

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| Dec 2026 | **PhD, Statistics**<br>University of Washington<br>Advisor: Alex Luedtke |
| May 2021 | **Bachelor of Science, Mathematics and Scientific Computing**<br>Indian Institute of Technology Kanpur<br>Advisor: Dootika Vats \| Minor: English Literature |

## Research Interests

Causal inference and semiparametric efficiency theory; optimal transport and distributional inference; uncertainty quantification and conformal prediction, particularly under distribution shift; gradient flows and mean-field dynamics of learning algorithms; machine learning for scientific applications, particularly ecology.

## Peer-Reviewed Publications

[1] **Medha Agarwal** and A. Luedtke. Sinkhorn treatment effects: A causal optimal transport measure. In *Forty-third International Conference on Machine Learning (ICML)*, 2026

[2] **Medha Agarwal**, K. Rafiq, R. Mehta, B. Abrahms, and Z. Harchaoui. Leveraging machine learning and accelerometry to classify animal behaviours with uncertainty. *Methods in Ecology and Evolution*, 17(2):567–584, 2026

[3] **Medha Agarwal**, Z. Harchaoui, G. Mulcahy, and S. Pal. Langevin diffusion approximation to same marginal Schrödinger bridge. *Journal of Functional Analysis*, page 111494, 2026

[4] **Medha Agarwal** and J. Xu. Quasi-Newton Acceleration of EM and MM Algorithms via Broyden's Method. *Journal of Computational and Graphical Statistics*, 2023

[5] **Medha Agarwal**, D. Vats, and V. Elvira. A principled stopping rule for importance sampling. *Electronic Journal of Statistics*, 2022

[6] **Medha Agarwal** and D. Vats. Globally centered autocovariances in MCMC. *Journal of Computational and Graphical Statistics*, 2022

## Preprints

[7] **Medha Agarwal**, Z. Harchaoui, G. Mulcahy, and S. Pal. Iterated Schrödinger bridge approximation to Wasserstein gradient flows. *arXiv preprint arXiv:2406.10823*, 2024

## Research Positions

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| Jun-Dec 2025 | **Abrahms Lab**, Research Intern, PI: Briana Abrahms, *Seattle, WA* |
| Mar-Jun 2024 | **eScience Institute**, Research Assistant, PI: Zaid Harchaoui & Briana Abrahms, *Seattle, WA* |
| 2023 | **Institute for Foundations of Data Science**, Research Assistant, PI: Zaid Harchaoui, *Seattle, WA* |
| Jul-Sept 2020 | **University of Edinburgh**, Research Intern, PI: Victor Elvira, *Edinburgh, Scotland* |
| May-Jul 2020 | **Duke University**, Research Intern, PI: Jason Xu, *Durham, NC* |

## Honors and Awards

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| 2026 | 2026 IMS International Conference on Statistics and Data Science, Student Travel Award *(declined)* |
| 2025 | Graduate Student Conference Presentation Award, University of Washington |
| 2024 | Winner, Student Paper Competition, ASA Conference on Statistical Learning and Data Science |
| 2023 | Hannan Graduate Student Award, Institute of Mathematical Statistics |
| 2022 | Center for Statistics and the Social Sciences Travel Award, University of Washington |
| 2022 | Scholarship recipient, Institute for Foundations of Data Science<br>*Supported by the NSF Transdisciplinary Research in Principles of Data Science (TRIPODS) program* |

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| 2021 | Department Fellowship, UW Statistics |
| 2021 | Proficiency Medal, Department of Mathematics and Statistics, IIT Kanpur<br>*Awarded for the best academic performance among graduating students in each department* |
| 2017 | Academic Excellence Award - Dr. Sangeeta Goel Memorial Award, IIT Kanpur<br>*Awarded for the highest womens' All India Rank (365) in IIT-Joint Entrance Exam* |

## Research Talks & Presentations

⋆ Invited talk • Oral presentation ◇ Poster

**Sinkhorn Treatment Effects: A Causal Optimal Transport Measure**

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| [1] Nov 2026 | ASA Section on Statistical Learning and Data Science Conference ⋆ | *Brooklyn, NY* |
| [2] Oct 2026 | Department of Health Care Policy, Harvard Medical School ⋆ | *Boston, MA* |
| [3] Jul 2026 | International Conference on Machine Learning (ICML) ◇ | *Seoul, South Korea* |
| [4] May 2026 | Optimal Transport + Optimization Workshop, NCCR SwissMAP • | *Les Diablerets, Switzerland* |

**Iterated Schrödinger Bridge Approximation to Heat Flow**

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| [5] Aug 2025 | Joint Statistical Meetings •<br>*Advances in Generative Models Session* | *Nashville, TN* |
| [6] Feb 2025 | Institute for Foundations of Machine Learning •<br>*Mathematics of Deep Learning Workshop* | *Austin, TX* |
| [7] Jan 2025 | Joint Mathematics Meetings ⋆<br>*Mathematics of Adversarial, Interpretable, and Explainable AI* | *Seattle, WA* |
| [8] Nov 2024 | ASA Conference on Statistical Learning and Data Science ◇ | *Newport Beach, CA* |
| [9] Oct 2024 | SIAM Mathematics of Data Science ◇ | *Atlanta, GA* |

**Leveraging Machine Learning and Accelerometry to Classify Animal Behaviors**

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| [10] Oct 2025 | Institute for Foundations of Data Science Seminar • | *Seattle, WA* |
| [11] Feb 2025 | UW Data Science Seminar ⋆<br>*Seminar Series for AI@UW Seed Grant Awardees* | *Seattle, WA* |

**Globally centered autocovariances in MCMC**

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| [12] May 2023 | BayesComp ⋆<br>*Session on Measuring Quality of MCMC Samples* | *Levi, Finland* |

## Selected Workshops Attended

† Invited attendee

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| Jul 2025 | **Wasserstein Gradient Flows in Math and ML Workshop** †<br>*Banff International Research Station* | *Banff, Canada* |
| Aug 2022 | **Deep Learning Theory Workshop and Summer School**<br>*Simons Institute for the Theory of Computing* | *Berkeley, CA* |
| Jul 2020 | **noRth 2020: A Virtual Conference for R Users** | *Minneapolis, MN* |

## Teaching Experience

**University of Washington**

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|---|---|
| Spring 2026 | STAT 498/591 - Special Topics: Optimization and Sampling |
| Winter 2026 | STAT 311 - Elements of Statistical Methods |
| Spring 2025 | CSE 546 - Machine Learning for Big Data |
| Autumn 2024 | STAT 512 - Statistical Inference I |
| Winter 2024, 2023 | STAT 513 - Statistical Inference II |
| 2021–2022 | STAT 311 - Elements of Statistical Methods |

**Indian Institute of Technology Kanpur**

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| Spring 2021 | MTH 511 - Statistical Simulation and Data Analysis |

## Industry Experience

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| Summer 2026 | **Quantbot Technologies**<br>Quantitative Researcher Intern | *New York City, NY* |
| | • Developing statistical learning methods for equity return prediction, with an emphasis on modeling idiosyncratic stock movements. | |
| Summer 2024 | **Amazon Science**<br>Applied Scientist Intern | *New York City, NY* |
| | • Developed multi-horizon probabilistic forecasting methods for multivariate time series using MQ-CNN; achieved $> 95\%$ empirical coverage on 5+ years of production-scale demand data. | |
| Summer 2023 | **Amazon Science**<br>Applied Scientist Intern | *Bellevue, WA* |
| | • Fine-tuned a 40B foundation model using reinforcement learning from human feedback, integrating human preferences with product-description corpora.<br>• Project paper featured in Amazon Machine Learning Conference 2023. | |

## Professional Service

**Reviewer**, *Journal of Statistical Planning and Inference*, 2026

## Academic Service

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| 2022-23 | Leader, Normalizing Flows Working Group (Webpage) | *UW Seattle* |
| Winter 2021 | Statistics Undergraduate Directed Reading Program (SPA-DRP) Mentor | *UW Seattle* |
| Autumn 2021 | Pre-application Review Service (PARS) volunteer | *UW Seattle* |
| 2019-20 | Student Undergraduate Committee Student Nominee | *IIT Kanpur* |